Coverage Report

GCD Module (UPC MIRI – PV)

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June 20, 2026

1 Methodology

Coverage is collected over three `uvm_tests` — *smoke* (bring-up), *directed* (corner scenarios), and *random* (constrained-random, 1000 transactions) — each run as an independent simulation. Every run is instrumented with QuestaSim code coverage (`+cover=bcefst`: statement, branch, condition, expression, FSM, toggle) and samples the functional covergroups in the coverage subscriber. Per-test databases are saved and combined with `vcover merge`; the merged database is the closure figure. In parallel, 14 `bind`-ed SVA assertions are checked every cycle.

2 Coverage Progression

Coverage is reported per test and merged, showing how the directed and random tests close the goals that the smoke test only partially reaches. Expression coverage is 100% in every run and is omitted from the table.

Test	Functional	Stmt	Branch	Cond	FSM St	FSM Tr	Toggle
smoke	44.14%	88.00%	81.81%	57.14%	100%	80.00%	25.09%
directed	81.05%	96.00%	90.90%	71.42%	100%	100%	100%
random	77.77%	96.00%	90.90%	71.42%	100%	80.00%	99.63%
merged	**100.00%**	**96.00%**	**90.90%**	**71.42%**	**100%**	**100%**	**100%**

Functional coverage closes to 100% on the merged database: the directed test supplies the reset-state and corner-value bins the random test cannot reach, while the random test fills the operand-value cross bins, so the two are complementary.

3 Final Code Coverage (merged) and Residual Gaps

Metric	Covered / Bins	Coverage
Statements	24 / 25	96.00%
Branches	20 / 22	90.90%
Conditions	5 / 7	71.42%
Expressions	2 / 2	100%
FSM states	3 / 3	100%
FSM transitions	5 / 5	100%
Toggles	271 / 271	100%

Every uncovered code item is **structurally unreachable**, so the reachable code coverage is 100%:

- `rtl/gcd.sv:118` — `default:  state <= IDLE;` (1 statement, 1 branch). The FSM is an exhaustively enumerated 3-state type, so the `default` arm can never execute. It is defensive coding only.
- Subtractive-step comparison false-legs (lines 51/53; remaining branch and condition misses, e.g. `(b_reg > a_reg)` false). In the `RUN` state the operands are always unequal, so the symmetric "false" leg of the comparison cascade is unreachable.
- Line 75 `(in_valid && in_ready)`, `in_ready` false term. This statement executes only in `IDLE`, where `in_ready` $\equiv 1$ by definition, so the term cannot be 0.

4 Closure

- **Functional coverage:** 100% of all covergroup bins (merged).
- **Code coverage:** 100% of reachable statements, branches, conditions, expressions, FSM states/transitions, and toggles; the only misses are the unreachable items listed above.
- **Assertions:** 14 / 14 exercised, 0 failures across all runs.
- **Tests:** smoke, directed, and random all pass with zero scoreboard value or latency mismatches.

The verification goals defined in the plan are met.